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Five Marks Question

1. In HDFS framework describe briefly the functionality of Task Tracker. Give a schematic diagram.
2. Volume, Variety, Velocity are the three major attributes of any big data. Give brief description of each of them.
3. A data set describing the crop yield in each year is given below;

Year = range(2020, 2025);

Apple = [4, 3, 9, 7, 6, 8]

xlabel = 'Year'

ylabel = 'tons per hector'

Title = 'Crop Yields'

Write the python program to produce the Bar graph of the given data.

4. Write the python code to draw a histogram of a given data set.
5. Write the python code to draw the scatter plot

6. An artificial neural network is described with the following parameters.

- i) No of i/p node = 3
- ii) No of o/p node = 1
- iii) The weights are given by —
 $W_1 = 2$, $W_2 = -4$ and $W_3 = 1$.

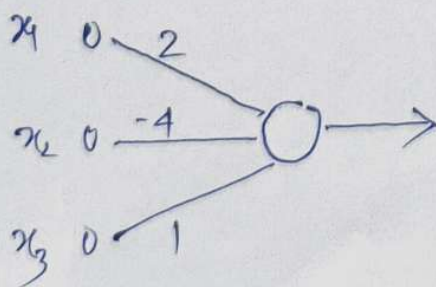
iv) The activation of the i/p nodes are unit while the activation of the o/p nodes is —

$$\phi(v) = 1 \text{ if } v \geq 0 \\ = 0 \text{ otherwise.}$$

$$v = \sum W_i x_i$$

Calculate the o/p of the network corresponding to the input pattern given below:

Pattern $\rightarrow$	P_1	P_2	P_3	P_4
x_1	1	0	1	1
x_2	0	1	0	1
x_3	0	1	1	1



$$v = 2 \times 1 + (-4 \times 0) + (1 \times 0) = 2$$

$\therefore v \geq 0 \therefore \phi(v) = 1$
 $\therefore \text{o/p} = 1$

$$v = 2 \times 0 + (-4 \times 1) + (1 \times 1)$$

$$= -4 + 1 = -3 < 0 \therefore \phi(v) = 0$$

7. A multilayer neural network has two input nodes, one hidden layer with two nodes and two output nodes. Draw the architecture of the network and show that each node of the hidden layer draws a straight line on the feature space.
8. Prove that in linear regression the least square line always passes through the mean values of x and y .

10 Marks Question

1. Derive the expression of the delta rule for a single neuron using back propagation and gradient descent.
2. Describe the Mapping operation in Map Reduce.
3. Describe with diagram the 'Name Node' and the 'data node' in Hadoop.
- 4.

- 4) In a class the number of hours of study and the corresponding pass (1) or Fail (0) history are given by a data set:

Hours of study : 15 28 29 33 39

Pass(1)/Fail(0): 0 0 1 1 1.

Based on the data a model is evolved relating the passing of the students and the hours of study as given below:

$$Z = -64 + 2 * \text{hours}.$$

Calculate the probability of passing for the students who studied for 32 hours.

- 5) Techmart is an online service provider. They conducted a survey and claims that individual customers pay on average Rs. 600 per month with a standard deviation of Rs. 25. A random sample of 50 Customer's bill during a given month is taken with a mean of Rs 550 and standard deviation Rs 15.

- (i) Identify the null hypothesis and the alternate hypothesis.
- (ii) Justify with proper explanation that the null hypothesis is accepted or rejected.

- 6) Briefly explain what is reinforcement learning using the example of designing a computer agent to play the Tic-Tac-Toe game.

- 7) Give a comparative study of Matplotlib and Seaborn.

- 8) Describe K-NN algorithm and give logic how to select the K values for a given problem.

- 9) Describe the secondary Name node and Job tracker in Hadoop.

- 10) In a feature plane the data are given by:

A1(2,10), A2(5,2), A3(8,4), A4(5,8), A5(7,5), A6(6,4), A7(1,2), A8(4,9).

The 8 feature points will be clustered into 3 groups. Run the k-means algorithm to cluster them. At the end of the first epoch (iteration) show:

- (i) New clusters.
- (ii) Centres of each new clusters.

- 11) Describe briefly the steps of the Map-reduce algorithm to multiply two matrices as given below:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 1 \\ 5 & 1 \end{bmatrix}.$$

MCQ

1. Which is true for neural networks
 - (a) Each node computes it's weighted input.
 - (b) Node could be in excited state or non-excited state.
 - (c) It has set of nodes and connections
 - (d) All of the above.
2. What is the objective of Backpropagation algorithm?
 - (a) To develop learning algorithm for multilayer neural network, so that network can be trained to capture the mapping implicitly.
 - (b) To develop learning algorithm for single layer feedforward neural network.
 - (c) Both (a) and (b).
 - (d) None of the above.
3. In a feed forward neural network with the following specifications:
 Input layer has three neurons and one output neuron. The input values are [0.3, 0.5, 0.6].
 The weights are [0.2, 0.1, -0.3]. The value of the pre activation to the output neuron is:
 (a) 1.7. (b) -0.07. (c) 0.07. (d) -1.7.
4. The learning algorithm of a multilayer perceptron is:
 - (a) Delta rule. (b) back propagation. (c) Gradient descent. (d) All of the above.
5. In sigmoid activation function the value of the function for the independent variable $x=0$ is:
 - (a) 0. (b) 1. (c) 0.5. (d) infinity.
6. To realize the two input XOR logic using multilayer perceptron the number of nodes in the hidden layer should be:
 - (a) 2. (b) 3. (c) 4. (d) None of the above.
7. For noisy data the value of 'K' in K-N-N algorithm must be:
 - (a) Small. (b) Large. (c) Medium. (d) None of the above.
8. The characteristics of Big data is commonly referred as :
 - (a) 2V. (b) 3V. (c) 4V. (d) None of the above.
9. The main computational frame work of Big data is:
 - (a) Map reduce. (b) HDLC. (c) Both of them. (d) None of the above.
10. Data visualization is:
 - (a) The process of collecting data. (b) The process of representing data graphically. (c) The process of cleaning data. (d) None of the above.